

Essential Goods

Let u be a *continuous* and *locally nonsatiated* real-valued function representing a preference relation $\succsim$ on the consumption set $X = \mathbb{R}_+^L$. For $(p, w) \gg 0$, define

$$d(p, w) = \{x \in B(p, w) \mid u(x) \geq u(y) \text{ for every } y \in B(p, w)\}$$

where $B(p, w) = \{x \in \mathbb{R}_+^L \mid p \cdot x \leq w\}$. Equivalently, $d(p, w)$ is the set of solutions to (UM).

Definition (Essentiality). *Good i is **essential** if $u(x) = u(0)$ whenever $x_i = 0$.*

Under LNS, if good i is essential, then the indifference set at any $\hat{x}$ with $\hat{x}_i > 0$ does not include any point with $x_i = 0$: in words, indifference sets with positive i consumption do not touch the $x_i = 0$ axis.¹

Fact. *If good i is essential and $x \in d(p, w)$, then $x_i > 0$.*

I prove the fact by contraposition. Let good i be essential, and let $x'_i = 0$ for some $x' \in B(p, w)$. I will show that $x' \notin d(p, w)$. By essentiality, $u(x') = u(0)$. And by LNS, for every $\varepsilon > 0$, there is a point $y \in \mathbb{R}_+^L$ with $y_\ell < \varepsilon$ for $\ell = 1, \dots, L$ with $u(y) > u(0) = u(x')$. Pick $\varepsilon > 0$ satisfying

$$\varepsilon < \frac{w}{\sum_{\ell}^L p_\ell},$$

so $y \in B(p, w)$. Since $u(y) > u(x')$, $x' \notin d(p, w)$. □

Here's an easy example. Suppose that $L = 2$ and that $u(x) = x_1 x_2$ represents a consumer's preferences. Clearly $u(0, x_2) = 0 = u(0)$ for every $x_2 \geq 0$, so that good 1 is essential (similarly, good 2 is also essential).

¹Recall that the indifference set at $x \in X$ is $I(x) = \{y \in X \mid y \sim x\}$.